

Project Definition: UniPathway

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1. Project Idea

UniPathway is a comprehensive full-stack web application designed to simplify and optimize the university admission process for prospective students. The platform acts as a smart academic consultant that allows users to input their grades (Bagrut, Psychometric, etc.), explore different academic departments across various universities, and receive AI-driven predictions regarding their chances of acceptance. The system is designed for high school graduates and pre-academic students navigating the complex admission requirements of the Israeli higher education system.

2. Project Purpose

The university application process is currently fragmented and stressful. Candidates must manually check different "Sekem" (weighted score) calculators for every institution, and they often lack clear insight into how likely they are to be accepted based on historical trends.

UniPathway solves this by:

- Centralizing admission data from multiple institutions.
- Automating the calculation of admission scores.
- Providing predictive insights to help users decide whether to retake exams or apply to alternative programs.
- Peer Insights: providing a platform for alumni to share their experiences, tips, and personal journeys

3. Main Components

- **Frontend:** The application Frontend provides a dynamic, responsive user interface, including interactive dashboards, progress trackers, and data visualization (charts) for admission trends.
- **Backend:** A REST API built with **Node.js and Express**. It handles the business logic, manages user authentication, performs complex score calculations, and interfaces with the AI models.
- **Database:** A **MySQL** relational database. It stores structured information about users, university departments, historical admission thresholds, and user-specific application lists.

4. Basic API design

Endpoint	Method	Purpose	Request Data (Frontend)	Response Data (Backend)
/api/auth/signup	POST	Creates a new user account.	{email, password, name}	{userId, token}
/api/user/grades	PUT	Updates user's academic scores.	{math_units, eng_units, psychometric, gpa}	{status: "success", updatedProfile}
/api/search/depts	GET	Filters departments by major or uni.	Query params: ?major=CS&city=Center	[[deptId, uniName, lastYearMinScore]]
/api/tracker	POST	Adds a department to user's watchlist.	{deptId}	{message: "Added to tracker"}

5. Pages / Screens

- **Login/Signup Page:** Secure entry point for users to create and manage their profiles.
- **User Dashboard:** The central hub where users see their "Quick Stats" (current scores) and a summary of their chances for saved departments.
- **Grade Profile Manager:** A dedicated form where users can update their Bagrut grades and Psychometric results to see real-time updates on their eligibility.
- **Department Explorer:** A searchable catalog where users can browse majors, view university details, and see "Match Scores" calculated by the system.
- **Comparison Tool:** A side-by-side view allowing users to compare 2-3 universities based on requirements, location, and AI-predicted acceptance difficulty.

6. Data

The system will store and process the following primary entities with MySQL:

- **User:** id, name, email, password_hash, psychometric_total, math_units, gpa, created_at.
- **University:** id, name, location, logo_url, website_url.
- **Department:** id, university_id, major_name, degree_type (B.Sc/B.A), faculty.
- **AdmissionThresholds:** id, department_id, year, min_score_required. (This historical data is used by the AI).
- **UserWatchlist:** id, user_id, department_id, status (Interested/Applied/Accepted)

7. AI integration ideas

- **Admission Probability Engine:** A machine learning model (or regression analysis) that predicts the likelihood of acceptance based on current grades vs. historical multi-year trends.
- **Personalized Recommendation System:** An algorithm that suggests alternative majors or institutions based on the user's existing profile and interests.
- **LLM Integration:** A language model (LLM) that will receive as input from the server both the user's question (for example: "I like biology but I'm bad at math") and the list of degrees they are already accepted to, in order to return realistic recommendations.

8. Basic Workflow

1. **Registration:** The user signs up and completes their academic profile by entering their grades.
2. **Discovery:** The user searches for a specific major (e.g., "Information Systems").
3. **Analysis:** The user views a specific department. The system automatically calculates their "Sekem" and triggers the **AI Component** to show the probability of acceptance.
4. **Tracking:** The user saves several options to their "Tracker" to monitor deadlines and status updates.
5. **Optimization:** The user uses the AI "What-If" tool to see which specific exam they should retake to maximize their chances.
6. **Decision:** Once the user reaches a high probability, they use the provided links to navigate to the official university registration portal.